Matthew Emmanuel T. Labrador

Software developer | Web applications, systems and applied AI

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EDUCATION

University of the Philippines Manila | BS Computer Science

Expected May 2029 | Cumulative GWA: 1.0375 (1.0 is highest) | DOST Undergraduate Scholar

CURRENT ROLES

NVIDIA Philippines - Campus Ambassador | 2026-present

PC Express - Intern | 2026-present

SELECTED ENGINEERING PROJECTS

Khanpanion | Developer & team lead

Next.js / React / TypeScript | Grand Champion, Khan Academy Education Impact Challenge 2026

- Developed a college-preparation platform with a typed prerequisite graph, generated practice and targeted Khan Academy resources.
- Built device-local learning state and resumable timed practice. Separated assisted attempts from fresh independent checks in the learning engine.

eGovMed | Full-stack development & service integrations

React / Express / Redis / eGovPH APIs | Grand Champion, eGov Hackathon PH 2026

- Connected eight government services across patient intake, identity, triage, booking and payment through explicit service adapters.
- Implemented mock/live boundaries and production validation. The public walkthrough preserves the patient UI with simulated services; PGH remains a pilot target.

Tumbang Preso | Sole developer

Godot / GDScript / ENet / Linux VPS | 1st Place, Gear Up NCR 2026

- Built the four-player competition game in five days: gameplay, bots, interface, physics, level assembly and multiplayer networking.
- Implemented host-authoritative networking, LAN discovery and dedicated online lobbies. Validated collision behavior with a 36-contact probe.

Knee MRI Reader | Data pipeline, model training & web application

Python / PyTorch / DINOv2 / ONNX / Next.js

- Built DICOM normalization and multi-view model training for 12 knee findings across 4,407 studies, then delivered a browser reading station.
- Evaluated the ensemble at 0.843 strict out-of-fold macro ROC-AUC. The deployed research viewer uses a single-member ONNX model.

ADDITIONAL TECHNICAL WORK

CHIP-8 - C++ interpreter, debugger and WebAssembly build; 35 instructions, 106 core assertions and six original ROMs.

GlycoSwarm AI - LangGraph workflow with four parallel specialists, a FastAPI service and Next.js frontend; preserved hackathon prototype.

TECHNICAL SKILLS

Languages: TypeScript, JavaScript, Python, C/C++, SQL and GDScript.

Web & systems: React, Next.js, Node.js, Express, FastAPI, Redis, WebAssembly, Git and Linux.

ML & data: PyTorch, scikit-learn, pandas, DINOv2, ONNX and LangGraph.